

KEVIN 'TYLER' COX

Software Engineer | Full-Stack, Platform & Game Systems

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[GitHub](#) | [LinkedIn](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Software engineer with 11+ years building Angular applications, backend integrations, delivery automation, and observability for logistics operations. Recently created CityRPG, a full-stack multiplayer roleplay platform spanning TypeScript, Angular 20/Firebase, and a .NET 8 Windows companion, and BMF, a modding framework built by reverse-engineering a proprietary Windows dedicated game server. Known for turning complex, stateful systems into testable, observable, operator-friendly products.

CORE TECHNOLOGIES

Frontend: Angular 20/22 | TypeScript | Signals | RxJS | NgRx | Angular Material | SSR | HTML/CSS

Backend & Data: Java/Spring Boot | C#/.NET 8 | Node.js | REST | Firebase/Firestore | Kafka/CDC | Python

Platform: GitHub Actions | GitLab Runners | Azure DevOps | Ansible | Kubernetes | GCP | Electron | Windows packaging

Observability & Security: Grafana | Prometheus | Loki | Tempo | SLOs & alerting | Keycloak | OIDC/SAML | PKI/TLS/mTLS

Game & Systems: Dedicated server reverse engineering | UE4SS | Lua | C++ integration | Rust | event-driven architecture | performance telemetry

PROFESSIONAL EXPERIENCE

Expeditors International

Aug 2014 - Present

Software Engineer / Angular Application Developer

Design and delivery of enterprise web applications and platform tooling for logistics operations.

- Built and maintained Angular front ends using Signals, RxJS, and NgRx across multiple logistics applications; established modular state patterns and reusable component boundaries.
- Integrated Angular applications with Java Spring Boot and C#/.NET APIs; developed typed clients and interceptors for authentication, telemetry, and consistent error handling.
- Implemented service observability with Prometheus metrics, Loki logs, Tempo traces, Grafana dashboards, SLOs, and alerting to improve detection and trace-led debugging.
- Standardized CI/CD with GitHub Actions and GitLab Runners; automated repeatable provisioning with Ansible and supported Kubernetes and GCP/Firebase deployments.
- Hardened identity and transport security through Keycloak SSO (OIDC/SAML), Kerberos integrations, internal PKI, TLS/mTLS, and certificate lifecycle controls.
- Modernized data integration from bespoke Kafka jobs to database-level Change Data Capture feeding Kafka with bronze, silver, and gold data layers.
- Enabled progressive delivery with feature flags and DORA metrics; coached teams on Kanban flow and automated Azure DevOps planning workflows with Python.

SELECTED ENGINEERING PROJECTS

CityRPG

Dec 2024 - Present

Creator & Lead Developer

Server-authoritative roleplay platform for a multiplayer sandbox game spanning a TypeScript server plugin, Firebase persistence, Rust world tooling, and native game-server integration.

- Architected server systems for economy, jobs, resources, properties, quests, player state, commands, and persistence, with feature flags and rollback-safe rollout paths.
- Implemented transactional Firestore workflows for trading, escrow, inventory drops, and anti-abuse policies with idempotent claims and server-owned state.
- Integrated server-authoritative BMF events, native resource-hit routing, socket commands, and bounded player-position data without relying on unsafe polling.
- Maintained 340+ automated test files, c8 coverage gates, Firebase emulator security-rule tests, and build/deployment automation across the monorepo.

SELECTED ENGINEERING PROJECTS (CONTINUED)

Reverse-Engineered Game Server Modding Framework (BMF)

Jun 2026 - Present

Creator & Maintainer

Server-side modding framework and operations stack created by reverse-engineering a commercial multiplayer video game's proprietary Windows dedicated server.

- Mapped engine/runtime objects, lifecycle callbacks, player and minigame data, native hook targets, and server control paths through runtime probes and live validation.
- Converted reverse-engineering findings into a UE4SS/Lua plugin runtime with lifecycle, storage, events, permissions, audit logging, and an authenticated socket bridge to a Node.js server manager.
- Developed native C++ hook and frame-telemetry modules plus Prometheus/Grafana instrumentation to attribute slow operations and enforce frame-time performance gates.
- Created an Angular 22/Electron 42 desktop console, CLI, and guarded orchestrator for install, repair, service control, telemetry, snapshots, rollback, and MSI/portable releases.
- Hardened runtime safety with bounded game-thread scheduling, capability gates, provenance checks, fail-closed player routing, and live canary validation.

CityRPG Web & Windows Companion

2025 - Present

Full-Stack & Desktop Developer

Angular/Firebase player portal and a non-injected .NET 8 WebView2 companion that brings game-linked workflows alongside the game client.

- Built an Angular 20 portal for profiles, economy, storefronts, quests, government, live activity, and a Wire-backed interactive city minimap.
- Developed a DPI-aware Windows companion with passive phone, interactive sidebar, and fullscreen modes; tracked game focus, window geometry, monitor changes, and customizable input shortcuts.
- Defined an origin-checked, versioned web protocol with no filesystem/process bridge and implemented signing-ready, versioned installers plus size- and SHA-256-verified self-updates.
- Automated Angular builds, Firebase security-rule and function tests, hosting deployment, release manifests, and guarded download publication.

CityRPG Economy & Progression Design

2025 - Present

Game Systems Designer

Data-driven economy and progression model for a persistent multiplayer city-roleplay server.

- Designed interlocking loops for jobs, resource gathering, delivery, property, player businesses, policing/crime, elections, and quest progression with explicit sources, sinks, prerequisites, and anti-farming controls.
- Created feature-flagged rollout plans, dependency-safe enable order, live validation gates, and recovery procedures for high-risk gameplay and native integration paths.
- Specified economy telemetry, balance metrics, seasonal and wipe models, and operating dashboards to tune progression and surface abuse without blocking legitimate play.

EDUCATION

Trident Technical College Computer Science

2013